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EDUCATION

- **Korea Advanced Institute of Science and Technology (KAIST)** *Daejeon, Korea*
  Ph.D. Candidate. CCS Graduate School of Mobility — Mar. 2023 - Feb. 2027
  M.S. CCS Graduate School of Mobility — Aug. 2020 - Feb. 2023
  Advisor: Prof. Namwoo Kang, (**Homepage**: smartdesignlab.org)
- **Korea University of Technology and Education** *Cheonan, Korea*
  B.S. Mechatronics Engineering — Feb. 2012 - Aug. 2019

EXPERIENCE

- **Graduate Researcher** *Daejeon, Korea*
  Smart Design Lab., KAIST — Sep. 2020 - Present
- **AI Researcher** *Daejeon, Korea*
  Narnia Labs — Feb. 2023 - Aug. 2024
- **Mechanical Design Engineer** *Incheon, Korea*
  ATI | Advanced Technology Inc. — May. 2019 - Oct. 2019
- **Student Researcher** *Cheonan, Korea*
  Computer-Aided-Engineering Lab., KOREATECH — Apr. 2017 - Dec. 2018

INDUSTRIAL PROJECTS

- **Samsung Heavy Industries**
  Generative AI-Based Sloshing Load Prediction and CCS Configuration Optimization — Apr. 2026 - Oct. 2026
- **Hyundai Motor Group**
  Development of a Battery Failure Prediction Algorithm Based on Anomaly Detection — Feb. 2024 - Mar. 2025
  Development of a GNN-Based Deep Learning Library for Collision Analysis Prediction — Sep. 2023 - Dec. 2023
  GNN-Based Deep Learning for Performance Prediction from 3D Geometry Data — Jul. 2022 - Jun. 2023
  Generative Design Guidelines and AI-Based Classification for Automotive Road Wheels — Jul. 2021 - Jul. 2022
- **LG Display**
  Optimization of Panel Layered Structures Using Generative Design and MeshGraphNet — May. 2022 - Feb. 2023
- **Korea Expressway Corporation**
  Development of an Expressway Accident Risk Index for Identifying Hotspots — Mar. 2021 - Dec. 2021
- **Korea Railroad Corporation (KORAIL)**
  Transportation Performance Analysis and Modal Substitutability under COVID-19 — Sep. 2020 - Feb. 2021

PUBLICATIONS

- **Park, J.**, & Kang, N.* (2026). "AutoATC: Automated Analytical Target Cascading Decomposition with LLM-Based Multi-Agent System." (Under review)
- **Park, J.**, & Kang, N.* (2026). "LLM-Guided Adaptive Penalty Parameter Updates for Analytical Target Cascading." (Under review)
- **Park, J.**, & Kang, N.* (2026). "Point-DeepONet: Predicting Nonlinear Fields on Non-Parametric Geometries under Variable Load Conditions." *Neural Networks, 108560.* (PDF) (code) (dataset)
- Kim, J., **Park, J.**, Kim, N., Yu, Y., Chang, K., Woo, C. S., Yang, S.*, & Kang, N.* (2025). "Physics-Constrained Graph Neural Networks for Spatio-Temporal Prediction of Drop Impact on OLED Display Panels." *Expert Systems with Applications, 126907.* (PDF)
- Hong, S., Kwon, Y., Shin, D., **Park, J.**, & Kang, N.* (2025). "DeepJEB: 3D Deep Learning-Based Synthetic Jet Engine Bracket Dataset." *Journal of Mechanical Design, 147(4).* (PDF) (dataset)
- **Park, J.**, & Kang, N. (2024). "BMO-GNN: Bayesian Mesh Optimization for Graph Neural Networks to Enhance Engineering Performance Prediction." *Journal of Computational Design and Engineering, 11(6), 260-271.* (PDF)
- Yun, J., Lee, J., **Park, J.**, Chung, K., & Lee, J.* (2022). "How to Measure the Network Vulnerability of Cities to Wildfires: Cases in California, USA." *Transportation research record, 2676(12), 382-395.* (PDF)

INTERNATIONAL CONFERENCES

- **Park, J.**, & Kang, N.* (2026). "LLM-Guided Adaptive Penalty Parameter Updates for Analytical Target Cascading", In *Asian Congress of Structural and Multidisciplinary Optimization (ACSMO 2026).*
- **Park, J.**, & Kang, N.* (2025). "A PointNet-Enhanced Operator Network for Nonlinear Analysis of Non-Parametric 3D Geometries Under Varying Load Conditions", In *16th World Congress of Structural and Multidisciplinary Optimisation (WCSMO 16).*
- Kim, J., **Park, J.**, Yang, S., & Kang, N.* (2024). "Design Optimization of Multi-layered Display Panels for Drop Impact using Physics-Constrained Graph Neural Network", *The 26th International Congress of Theoretical and Applied Mechanics 2024 (ICTAM2024).*
- **Park, J.**, & Kang, N.* (2023). "Bayesian Mesh Optimization for Graph Neural Networks to Enhance Engineering Performance Prediction", In *ASME 2023 International Design Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC/CIE 2023).*
- **Park, J.**, & Kang, N.* (2023). "How to Apply AI to Real-World Design Problems", In *ASME 2023 International Design Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC/CIE 2023).*
- Yun, J., Lee, J.Y., **Park, J.**, Chung, K., and Lee, J.* (2022), "How to Measure the Vulnerability of Cities to Wildfires: Cases in California", *Transportation Research Board (TRB) 101st Annual Meeting.*

DOMESTIC CONFERENCES

- Cho, Y., **Park, J.**, & Kang, N.* (2026). "A Data-Driven Framework for Railway Track Condition Assessment Using the Analytic Hierarchy Process", In *The Korean Society of Mechanical Engineers (KSME).*
- **Park, J.**, & Kang, N.* (2025). "DialogCAD: Interactive CAD Generation Framework with a Multi-Agent System via Model Context Protocol", In *The Korean Society of Mechanical Engineers (KSME).*
- **Park, J.**, Kim, J., Jang, W., Lee, Y., Lee, S., & Kang, N.* (2025). "A Study on 3D Topology Optimization Shape Reconstruction with CSG-Based Deep Learning", In *The Korean Society of Mechanical Engineers (KSME).*
- **Park, J.**, & Kang, N.* (2025). "SG-DeepONet: A Spatial Gradient-Aware Deep Operator Network for Pressure and Shear Stress Prediction in Vehicle Aerodynamic Analysis", In *The Korean Society of Mechanical Engineers (KSME).*
- **Park, J.**, & Kang, N.* (2024). "Stress Field Prediction in Nonlinear Static Analysis for Arbitrary 3D Geometries Using Latent-DeepONet", In *The Korean Society of Mechanical Engineers (KSME).*
- Hong, S., Kwon, Y., Shin, D., **Park, J.**, and Kang, N.* (2024). "DeepJEB: 3D Deep Learning-based Synthetic Jet Engine Bracket Dataset", In *The Korean Society of Mechanical Engineers (KSME).*
- Shin, D., **Park, J.**, Seo, M., & Kang, N.* (2024). "3D Field Prediction and Uncertainty Quantification with Implicit Neural Representation", In *The Korean Society of Mechanical Engineers (KSME), 289-290.*
- **Park, J.**, Kim, J., Yu, Y., Kim, N., Ha, S., Jang, K., & Kang, N.* (2023). "A Study on Impact Analysis based on Graph Neural Networks for Optimization of the Display Panel Structure", In *The Korean Society of Mechanical Engineers (KSME), 144-145.*
- **Park, J.**, & Kang, N.* (2022). "Bayesian Mesh Optimization for Graph Neural Networks", In *The Korean Society of Mechanical Engineers (KSME), 1626-1628.*
- **Park, J.**, Shin, D., Yoo, S., & Kang, N.* (2022). "Preliminary Study on 3D Wheel Performance Prediction Using Graph Neural Networks", In *The Korean Society of Mechanical Engineers (KSME), 316-317.* (Best Paper)
- Yun, J., Lee, J., **Park, J.**, Chung, K., & Lee, J.* (2021). "How to Measure the Vulnerability of Cities to Wildfires, Cases in California", In *Korean Society of Transportation, 436-437.*
- Mok, H, **Park, J.**, Lee, J., Han, D., & Kim, D.* (2021). "Development of Express Accident Risk Index (ex-ARI) for Identifying Hotspots", In *The Korean Society of Civil Engineers (KSCE), 57-58.*
- Mok, H., Lee, J., Han, D., **Park, J.**, & Kim, D.* (2021). "Establishment of Traffic Safety Measures based on Accident Risk Idex(ex-ARI), Cases in Dangjin Yeongdeok Expressway", In *The the Korean Society of Civil Engineers (KSCE), 506-507.*
- **Park, J.**, Kim, D., Mok, H., Han, D., & Lee, J.* (2021). "Identifying Fatal and Severe Accident Hotspots based on Multiple Safety Performance Measures", In *The Korean Society of Civil Engineers (KSCE), 49-50.* (Best Paper)

AWARDS

- **Best Presentation Award**, Korean Society of Mechanical Engineers — May 2022
- **Best Presentation Award**, Korean Society of Civil Engineering — Nov 2021
- **Best Poster Award**, Korean Society of Civil Engineering — Nov 2021
- **Minister of Science, Technology & Telecommunication Award**, Capstone Design Fair — Oct 2018

- **Korea Institute for Robot Industry Advancement Award**, SEOULTECH Robot Competition Oct 2018
- **Chancellor's Award, Kyungnam University**, Changwon National Robot Competition Sep 2018
- **KOREATECH Industrial-Academic Cooperation Award**, SolidWorks Technology Competition Dec 2017
- **Chancellor's Award, KOREATECH**, University Engineering Design Contest Nov 2017

OTHERS

- **TA**, KAIST MO833 Lecture: "AI-based Mobility Design", Overview of DeepONet Applications Nov. 2024
- **TA**, KAIST MO833 Lecture: "AI-based Mobility Design" Mar. 2024 - Jun. 2024
- **TA**, KAIST Industry-Academic Collaboration Lecture: "Analysis Prediction to Optimization with AI" Feb 2024
- **Patent**, Apparatus and Method for Interactive CAD Generation Based on a Multi-Agent System via a Model Context Protocol, KR 10-2025-0091992 May 2025
- **Patent**, Wheelchair Maneuverability Enhancement, KR 10-2018-0015617 Feb 2018